

Minimal Subsampled Rank-1 Lattices for Multivariate Approximation with Optimal Convergence Rate

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UNSW
SYDNEY

Worst-case setting

Given:

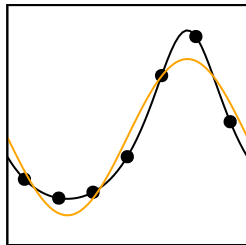
- $L_2 = L_2(\mathbb{T}^d, d\mathbf{x})$
- class of functions F

Goal: find

- points $\mathbf{X} = \{\mathbf{x}^1, \dots, \mathbf{x}^n\} \subset \mathbb{T}^d$
- efficient algorithm $S^{\mathbf{X}}$

with small **worst-case error**

$$\sup_{f \in F} \|f - S^{\mathbf{X}} f\|_{L_2}$$



- f
- $(\mathbf{x}^i, f(\mathbf{x}^i))$
- $S^{\mathbf{X}} f$

Korobov spaces

Definition

The **weighted Korobov space** is given by

$$H = H_{d,\alpha,\gamma} := \left\{ f: \mathbb{T}^d \rightarrow \mathbb{C} : \|f\|_H^2 = \sum_{\mathbf{h} \in \mathbb{Z}^d} r(\mathbf{h}) |\hat{f}_{\mathbf{h}}|^2 < \infty \right\},$$

with $r(\mathbf{h}) = r_{d,\alpha,\gamma}(\mathbf{h}) := \gamma_{\text{supp}(\mathbf{h})}^{-1} \prod_{j \in \text{supp}(\mathbf{h})} |h_j|^{2\alpha}$.

special cases include:

- $\gamma_{\mathbf{u}} = 1$ (unweighted) Sobolev spaces of dominating mixed smoothness
- $\gamma_{\mathbf{u}} = \prod_{j \in \mathbf{u}} \gamma_j$ product weights
- $\gamma_{\mathbf{u}} = \Gamma_{|\mathbf{u}|}$ order-dependent weights
- $\gamma_{\mathbf{u}} = \Gamma_{|\mathbf{u}|} \prod_{j \in \mathbf{u}} \gamma_j$ POD weights
- $\gamma_{\mathbf{u}} = \sum_{\boldsymbol{\nu}_{\mathbf{u}} \in \{1, \dots, \alpha\}^{|\mathbf{u}|}} \Gamma_{\|\boldsymbol{\nu}_{\mathbf{u}}\|_1} \prod_{j \in \mathbf{u}} \gamma_{j, \nu_j}$ SPOD weights

Algorithms

Least squares approximation

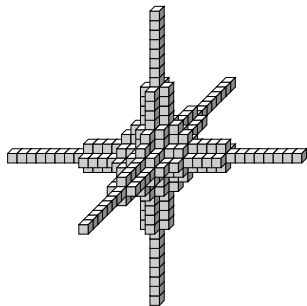
input: • samples $X \subset \mathbb{T}^d, g(x^i) \in \mathbb{C}$
• frequencies $\mathcal{B} \subset \mathbb{Z}^d$

method: $V := \text{span}\{\exp(2\pi i \langle \mathbf{h}, \cdot \rangle)\}_{\mathbf{h} \in \mathcal{B}}$

$$\begin{aligned} S_{\mathcal{B}}^X f &:= \arg \min_{g \in V} \sum_{i=1}^n |f(x^i) - g(x^i)|^2 \\ &= \sum_{\mathbf{h} \in \mathcal{B}} \hat{g}_{\mathbf{h}} \exp(2\pi i \langle \mathbf{h}, \cdot \rangle) \end{aligned}$$

computation: $\hat{g} = (L^* L)^{-1} L^* \mathbf{f}$ with

$$L = [\exp(2\pi i \langle \mathbf{h}, \mathbf{x}^i \rangle)]_{i \in \{1, \dots, n\}, \mathbf{h} \in \mathcal{B}}$$



hyperbolic cross frequency index set

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Kernel method

input: • samples $X \subset \mathbb{T}^d, f(x^i) \in \mathbb{C}$
• kernel $K: \mathbb{T}^d \times \mathbb{T}^d \rightarrow \mathbb{C}$

method:

$$A_{\text{ker}}^X f := \sum_{i=1}^n a_i K(\cdot, x^i)$$

with $a = K^{-1} f$ and

$$K = [K(x^i, x^j)]_{i, j \in \{1, \dots, n\}}$$

feature: given points X , this method has the best worst-case error

Points

Definition: Rank-1 lattice

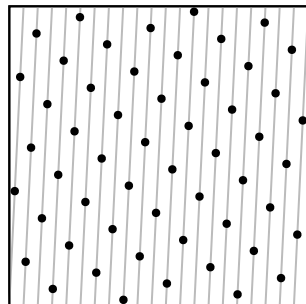
$$\mathbf{X} = \left\{ \frac{k}{n} \mathbf{z} \bmod 1 \right\}_{k=0}^{n-1} \subset \mathbb{T}^d$$

with **generating vector** $\mathbf{z}$ and **lattice size** n .

Component-by-component (CBC) algorithm

```
 $z_1 := 1$   
for  $j = 2, \dots, d$   
  choose  $z_j$  based on criterion  
end
```

- FFT applicable



$$\mathbf{z} = (1, 21)^\top, n = 55$$

Theorem

[Kämmerer, Potts, Volkmer 2015]

Constructing z by CBC fulfilling the
reconstructing property

$$\langle \mathbf{h}, z \rangle \not\equiv_n \langle \ell, z \rangle \quad \forall \mathbf{h} \neq \ell \in \mathcal{B}$$

yields for smoothness $\alpha > 1/2$ and $\varepsilon > 0$

$$\sup_{\|f\|_H \leq 1} \|f - S_{\mathcal{B}}^{\mathbf{X}} f\|_{L_2} \lesssim n^{-\alpha/2+\varepsilon}.$$

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Theorem

[Cools, Kuo, Nuyens, Sloan 2020]

Constructing z by CBC minimizing

$$\mathcal{S}_n(z) = \sum_{\mathbf{h} \in \mathbb{Z}^d} \frac{1}{r(\mathbf{h})} \sum_{\substack{\ell \in \mathbb{Z}^d \setminus \{\mathbf{0}\} \\ \langle \ell, z \rangle \equiv_n 0}} \frac{1}{r(\mathbf{h} + \ell)}$$

yields for smoothness $\alpha > 1/2$ and $\varepsilon > 0$

$$\sup_{\|f\|_H \leq 1} \|f - S_A^{\mathbf{X}} f\|_{L_2} \lesssim n^{-\alpha/2+\varepsilon}.$$

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Theorem

[Byrenheid, Kämmerer, T. Ullrich, Volkmer 2017]

Any algorithm using lattice points satisfies

$$\sup_{\|f\|_H \leq 1} \|f - S^{\mathbf{X}} f\|_{L_2} \gtrsim n^{-\alpha/2}$$

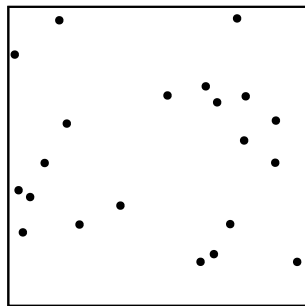
Theorem

[Krieg, M. Ullrich 2019]

Let X be n i.i.d. uniform random points.
Then for $\varepsilon > 0$

$$\sup_{\|f\|_H \leq 1} \|f - S_B^X f\|_{L_2} \lesssim \left(\frac{n}{\log n} \right)^{-\alpha + \varepsilon}.$$

- there is a logarithmic gap, which was closed by
[Nagel, Schäfer, T. Ullrich 2022], [Dolbeault, Krieg, M. Ullrich 2022],
[B, Schäfer, T. Ullrich 2023]
- no fast algorithm due to lack of structure

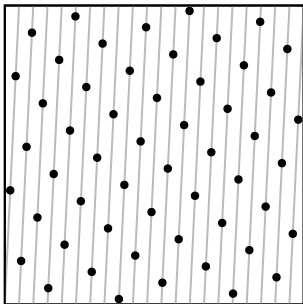


random points

Subsampled rank-1 lattices

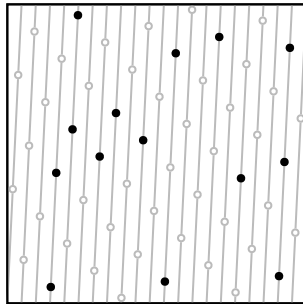
1

initial rank-1 lattice X



2

randomly subsampled X_J



Fast algorithm

- initial rank-1 lattice $\mathbf{X} = \{\frac{k}{n}\mathbf{z} \bmod 1\}_{k=0}^{n-1}$ with $\mathcal{O}(n \log n)$ algorithm
- small point set $\mathbf{X}_J = \{\frac{k}{n}\mathbf{z} \bmod 1\}_{k \in J} \subset \mathbf{X}$

Fast algorithm reusable

We have • $\mathbf{L}_J = \mathbf{P}\mathbf{L}$ for the least-squares matrix and

- $\mathbf{K}_J = \mathbf{P}\mathbf{K}\mathbf{P}^*$ for the kernel matrix with

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 & 0 \\ & \vdots & & & & \vdots & \\ 0 & 0 & 0 & \dots & 0 & 1 & 0 \end{pmatrix} \begin{matrix} 1 \\ 2 \\ \vdots \\ n \end{matrix}$$

$j_1 \qquad \qquad j_2 \qquad \qquad \qquad j_n$

- with $|J| \sim \sqrt{n} \log n$ the naive and FFT-approach have the same arithmetic cost

Theoretical guarantees

Theorem

[B, Kämmerer, Potts, T. Ullrich 2022]

For unweighted Korobov spaces with $\alpha > 1/2$ there is a subsampled lattice $\mathbf{X}_J$ with

$$\sup_{\|f\|_H \leq 1} \|f - S_{\mathbf{B}}^{\mathbf{X}_J} f\|_{L_2} \lesssim |J|^{-\alpha} (\log |J|)^{(d-1)\alpha+1/2},$$

and initial lattice size

$$n \sim |J|^{\frac{2}{1-1/(2\alpha)}}.$$

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Theorem

[B, Gilbert, Kuo, Sloan 2025]

For Korobov spaces with $\alpha > 1/2$ there is a subsampled lattice $\mathbf{X}_J$ with $\varepsilon > 0$

$$\sup_{\|f\|_H \leq 1} \|f - S_{\mathbf{B}}^{\mathbf{X}_J} f\|_{L_2} \lesssim \left(\frac{|J|}{\log |J|} \right)^{-\alpha+\varepsilon},$$

and initial lattice size

$$n \lesssim \left(\frac{|J|}{\log |J|} \right)^{\frac{2}{\sqrt{1-\varepsilon/\alpha}}}.$$

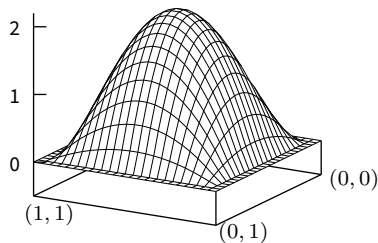
Note, rate $|J|^{-\alpha+\varepsilon}$ implies $n \gtrsim |J|^{2-\frac{2\varepsilon}{\alpha}}$.

First numerical experiment

- example function in $d = 2$

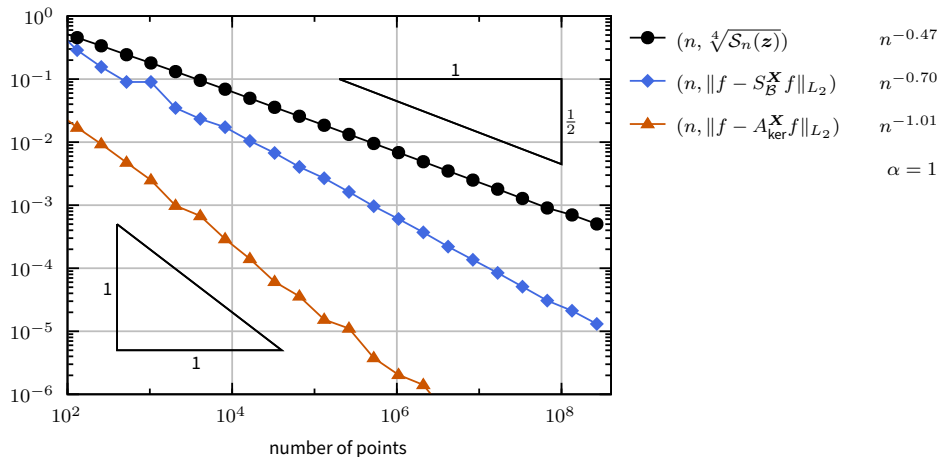
$$f(\mathbf{x}) = \left(\frac{5^{3/4}15}{4\sqrt{3}}\right)^d \prod_{j=1}^d \max\left\{0, \frac{1}{5} - \left(x_j - \frac{1}{2}\right)^2\right\}$$

- f has smoothness $3/2 - \varepsilon$
- rank-1 lattice from [Gilbert, Sloan WIP]
- least-squares approximation and kernel method
- full and subsampled rank-1 lattice



First numerical experiment

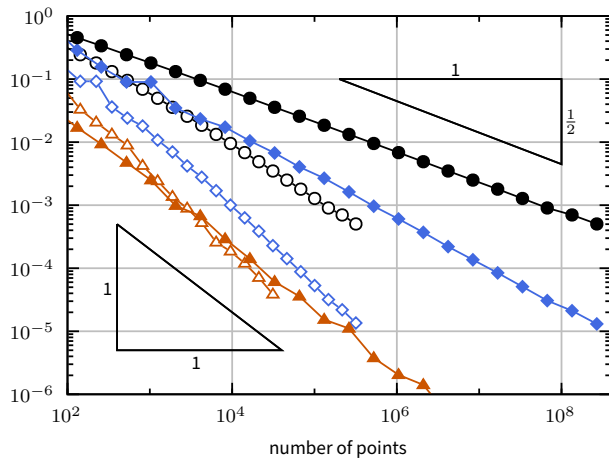
kink function $d = 2$



- [Kaarnioja, Sloan 2025] on doubling the rate

First numerical experiment

kink function $d = 2$



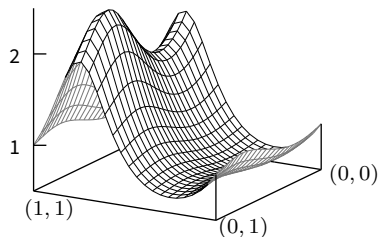
$\bullet$	$(n, \sqrt[4]{\mathcal{S}_n(z)})$	$n^{-0.47}$
$\blacklozenge$	$(n, \ f - S_B^X f\ _{L_2})$	$n^{-0.70}$
$\blacktriangle$	$(n, \ f - A_{\text{ker}}^X f\ _{L_2})$	$n^{-1.01}$
$\circ$	$(J , \sqrt[4]{\mathcal{S}_n(z)})$	$\left(\frac{ J }{\log J }\right)^{-0.93}$
$\blacklozenge$	$(J , \ f - S_B^{X_J} f\ _{L_2})$	$\left(\frac{ J }{\log J }\right)^{-1.37}$
$\blacktriangle$	$(J , \ f - A_{\text{ker}}^{X_J} f\ _{L_2})$	$\left(\frac{ J }{\log J }\right)^{-1.47}$
		$\alpha = 1$

Second numerical experiment

- analytic example function in $d = 100$

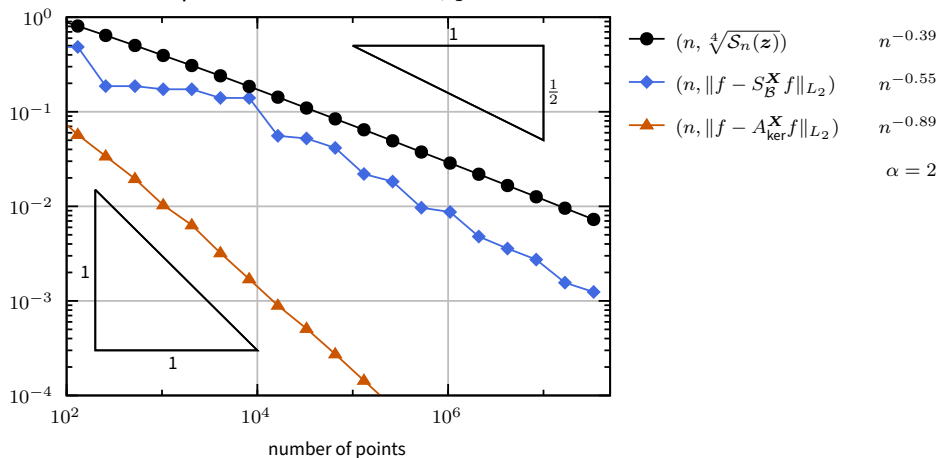
$$f(\mathbf{x}) = \frac{1}{a(\mathbf{x})}, \quad a(\mathbf{x}) = 1 + \frac{1}{2} \sum_{j=1}^d j^{-2.5} \sin(2\pi x_j)$$

- rank-1 lattice from [Gilbert, Sloan wip]
- least-squares approximation and kernel method
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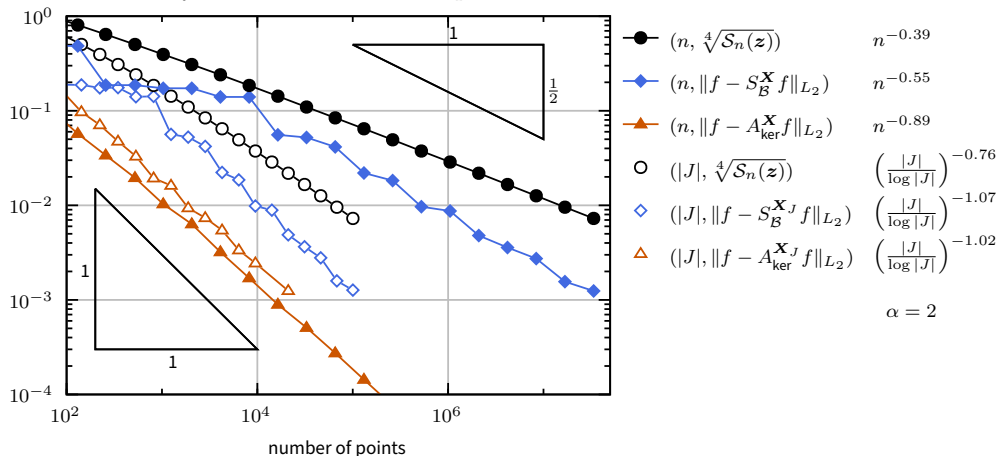
Second numerical experiment

reciprocal function $d = 100, q = 2.5$

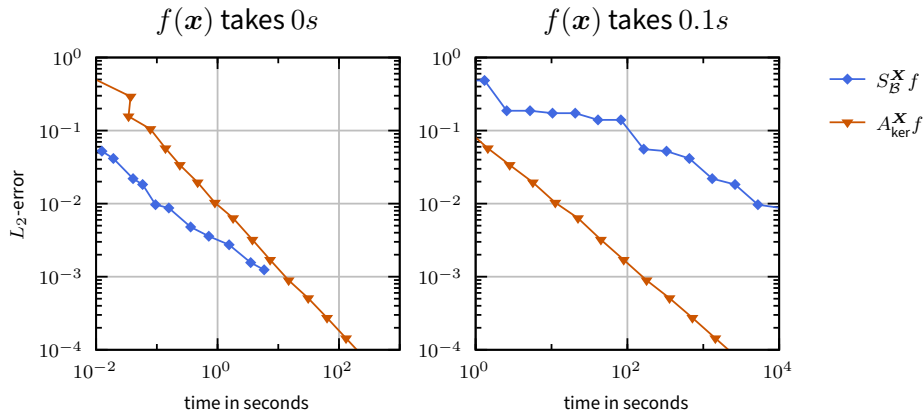


Second numerical experiment

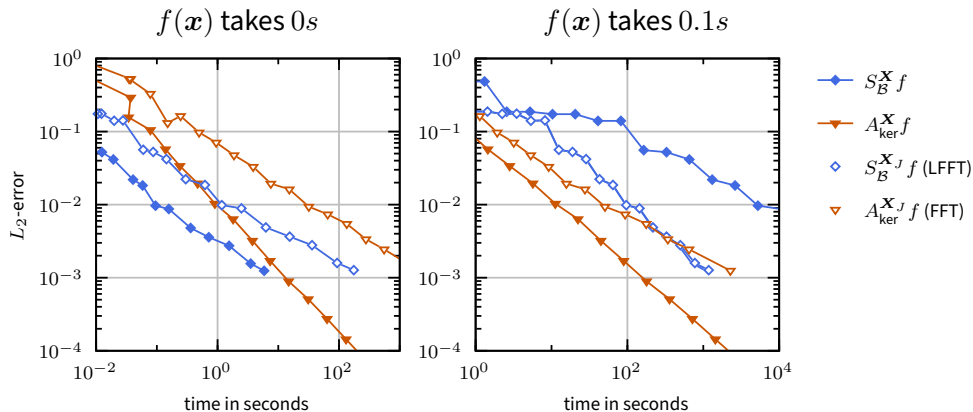
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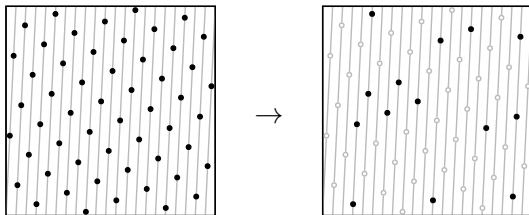
Second numerical experiment



Second numerical experiment



Conclusion



- we propose a **random subsampling** procedure for **rank-1 lattices**
 - achieving near optimal sampling complexity
 - making it possible to utilize FFT algorithms
 - with minimal initial rank-1 lattice size
- [B, Gilbert, Kuo, Sloan] on **arXiv:2506.07729**

Extra: connection of the CBC approaches

Theorem

[B, Gilbert, Kuo, Sloan 2025]

Let • $\mathbf{X} = \{\frac{k}{n}z \bmod 1\}_{k=0}^{n-1} \subset \mathbb{T}^d$ and
• $A^{\mathbf{X}}$ an approximation algorithm using samples of $\mathbf{X}$.

Then $\mathbf{X}$ has the **reconstructing property**

$$\langle \mathbf{h}, z \rangle \not\equiv_n \langle \ell, z \rangle \quad \forall \mathbf{h}, \ell \in \mathcal{B}$$

for

$$\left\{ \mathbf{h} \in \mathbb{Z}^d : r(\mathbf{h}) < \left(\sup_{\|f\|_H \leq 1} \|f - A^{\mathbf{X}}\|_{L_2} \right)^{-2} \right\}.$$